

SUMMARY

- Data Scientist with experience in ETL pipelines, predictive modeling and AI. Adept in translating advanced statistical and ML methods into actionable insights through interactive dashboards (Tableau, Power BI, Python visualization), supporting data-driven decision making for healthcare and enterprise stakeholders.

WORK EXPERIENCE

- **Georgetown University** Washington, USA
Data Scientist Sep 2024 – Dec 2024
 - Built a **ETL pipeline using Python and PostgreSQL**, containerized with Docker, to streamline ingestion of more than 30 GB of unstructured EHR data, reducing processing time by 60%.
 - **Fine-tuned Med-BERT and BEHRT models** on EHR data for **Type-2 diabetes prediction**, achieving 80% classification accuracy and enabling earlier clinical decision support.
 - Validated **transformer-based models** outperformed traditional baselines (**logistic regression, LSTMs**) for Type-2 diabetes onset prediction, improving predictive accuracy and supporting proactive clinical interventions.
- **Medequip, Inc** Irvine, California
Data Engineer Jun 2024 – Aug 2024
 - Utilized **LLMs with prompt engineering** and template-based techniques to convert unstructured clinical notes and claims information into structured reports.
 - Built scalable **Automated ingestion of this structured data** into Medisoft software, reducing manual data entry.
- **Department of Health Management - Georgetown University** Washington, USA
Data Scientist November 2023 – Jan 2024
 - Performed **Data Visualizations in Python** to understand discrepancies in vaccine waste, **Medicare** plan distributions in CMS.
 - **Utilized KMeans Clustering** to strategically identify and differentiate regions with high and low vaccine wastage, facilitating targeted intervention strategies for efficient vaccine distribution.
- **Micron Technologies, LLC** Hyderabad, India
Software Developer Jun 2021 – Feb 2023
 - Reduced HR data processing time by 87.5% (6 hours to 45 minutes) through **MySQL partitioning strategies** and stored procedure optimization, enabling same-day payroll validation for 40,000+ employees
 - Eliminated 40+ hours of weekly manual work by **architecting UiPath RPA workflows** for Workday report extraction and document generation, achieving 98% automation success rate
 - Built serverless **ETL pipelines using AWS Glue and Lambda**, processing 10K+ employee records daily with 35% throughput improvement and 30% infrastructure cost reduction
 - Accelerated HR decision-making by 80% through **Tableau dashboard** implementation, providing real-time visibility into workforce metrics previously available only through day-old batch reports

PROJECTS

- **Healthcare Cost Prediction and Analytics System** — *Regression, Regularization, Tableau, healthcare*
 - Built **machine learning models (Linear/Ridge Regression, Gradient Boosting)** with stratified 5-fold cross-validation, applying hyperparameter tuning, and regularization, in predicting hospitalization costs
 - Delivered actionable insights through interactive **Tableau dashboards** enabling data-driven decision support that reduced manual analysis time by 40% for healthcare stakeholders.
- **Multi-Agent AI Deal Detection System** — *LLM, ChromaDB, Ensemble Learning, Linear Regression, BeautifulSoup*
 - Engineered multi-agent AI system using **LLMs, RAG, ensemble learning** to detect and price product deals.
 - Optimized **ChromaDB with 400,000+ product embeddings** using sentence transformers to get pricing data.
 - Developed an ensemble pricing system combining **LLMs and Random Forest models** with weighted linear regression, improving price prediction accuracy by 23%.

- Architected an automated system monitoring RSS feeds and processing product descriptions using **NLP**.
- **GenAI-Powered Data Engineering Agent** — *LangChain, OpenAI, Streamlit, Docker, AWS ECS, FastAPI*
 - Designed **multi-agent orchestration using LangGraph(OpenAI, LangChain)**, enabling task delegation between specialized LLM agents reducing error propagation by 30%.
 - Implemented **ChromaDB as a vector store** for retrieval of data preprocessing and Data Engineering best practices for context-aware LLM responses
 - Built a **Streamlit-based UI for real-time monitoring of ETL jobs** and interactive query resolution, empowering non-technical users to interact with GenAI-powered pipelines.
 - Integrated **FastAPI endpoints** for external consumption of GenAI services, allowing seamless connection with downstream applications and **RESTful automation**

EDUCATION

- **Georgetown University** Washington, USA
Master of Science in Data Science and Analytics — GPA: 3.95 *Aug 2023 – May 2025*
- **Chaitanya Bharathi Institute of Technology** Hyderabad, India
Bachelors in Computer Science — GPA: 3.7 *Aug 2017 – May 2021*

TECHNICAL SKILLS

- **Data Engineering** : SQL (MySQL, PostgreSQL, Snowflake), Python (Pandas, NumPy), Data Integrity & Governance, ETL Pipelines, Docker, Airflow, DBT, AWS (Lambda, ECS, Glue)
- **Statistical & ML Techniques** : Regression, Survival Analysis, Time Series, Predictive Modeling, Root-Cause Analysis, A/B Testing, Trend Analysis, Forecasting
- **Visualization & Reporting** : Power BI, Tableau, VBA macros, Matplotlib, Plotly, Seaborn
- **Frameworks & Tools** : FastAPI, Flask, Django, Git, CI/CD (GitHub Actions), Kubernetes, Terraform, Spark